5 Interrupts

Introduction

The 8085 interrupt process is controlled by the interrupt enable flip-flop, which is internal to the processor and can be set or rest by using software instructions. If the flip-flop is enabled and the input to the interrupt signal INTR goes high, the microprocessor is interrupted. This is a maskable interrupt and can be disabled. The 8085 has a non-maskable interrupt and three additional vectored interrupt signals as well. In effect, the method would provide an external asynchronous input that would inform the processor that it should complete the instruction currently being executed and fetch a new routine that will serve the requesting device. Once this servicing is completed, the processor would resume exactly where it left OFF. This method is called the interrupt method.

Types of Interrupt

The interrupts are classified as follows.

1. Hardware interrupt
2. Software interrupt

Software Interrupts

When any of the restart instructions is included in the main program as software instruction, the instruction calls the subroutines stored at addresses corresponding to these instructions. This one byte call of subroutine by restart instruction is called a software interrupt. The 8085 has eight software interrupts from RST0 to RST 7.

Hardware Interrupts

When an interrupt service routine is called due to an interrupt request from external hardware, it is called hardware interrupt. INTR, RST 5.5, RST 6.5, RST 7.5 and TRAP are examples of hardware interrupts.

Vectored and Non-vectored Interrupts

Vectored interrupts: these interrupts are automatically vectored (transferred) to specific locations on memory page 00H without any external hardware. RST 5.5, RST 6.5, RST 7.5, and TRAP are the four vectored interrupts of 8085. The addresses assigned to the vectored interrupts are TRAP – 0024H

Non-vectored interrupts: INTR is the only non-vectored interrupt of 8085. When INTR interrupt is given to the microprocessor, it completes the execution of current instruction being executed, and then, generates the interrupt acknowledgment signal called $\overline{\text{INTA}}$.

Maskable and Non-Maskable Interrupts

Maskable interrupts: INTR, RST 5.5, RST 6.5, RST 7.5 are maskable interrupts of 8085. The maskable interrupts can be enabled using the instructions EI in the program. These interrupts can be disabled either by calling the subroutine corresponding to them or by giving software instruction like DI and SIM.

Note: INTR can be disabled using instruction DI only, whereas RST 5.5, RST 5.5, RST 7.5, can be disabled by using either DI or SIM instructions.

Note: SIM is the additional control for vectored maskable interrupt.

Non-maskable interrupts: TRAP is the only non-maskable interrupt of 8085 microprocessors. It cannot be disabled by any instruction or subroutine. When signal or TRAP pin is made high, the microprocessor stops the execution of current instruction being executed and

immediately calls the subroutine stored at 0024H without any external hardware or interrupt enable instruction EI.

Priority Levels of Interrupts

Trap	Highest priority
RST 7.5	↑
RST 6.5	↑
RST 5.5	↑
INTR	Lowest priority

Note: Hold signal has higher TRAP and an interrupt can be recognised only if the HOLD signal is inactive. Reset has higher priority than 7.5, 6.5, 5.5, INTR.

Note:

1. RST 5.5 and RST 6.5, INTR are level triggered. The signal should be kept high until the microprocessor completes the execution of current instruction.
2. RST 7.5 is positive edge triggered and can be triggered by a short pulse. The request is stored by D flip-flop internally until either the microprocessor responds to the request or it is cleared by reset or SIM instruction.
3. TRAP is edge, as well as, the level triggered means by which the input should go high and stay high to be acknowledged. It cannot be acknowledged again until it makes a transition from high to low to high.

Instructions Related to Interrupts

1. EI
   a. Enables all maskable interrupts
   b. 1-byte instruction
   c. No addressing mode
   d. One machine cycles (OP) and 4 T-states
   e. No flag is affected.
2. DI
   a. Disable all maskable interrupts
   b. 1-byte instruction
   c. No addressing mode
   d. One machine cycles (OP) and 4 T-states
   e. No flag is affected.
3. SIM
   a. It stands for Set Interrupt Mask. It masks the vectored maskable interrupts individually and transfers data serially to the SOD pin.
   b. 1-bytes instruction
   c. No addressing mode
   d. One machine cycles (OP) and 4 T-states
   e. No flag is affected.
4. RIM
   a. It stands for Read Interrupt Mask. The accumulator is loaded automatically with contents depending on the status of various interrupts and SID pin.
   b. 1-bytes instruction
   c. No addressing mode
   d. One machine cycles (OP) and 4 T-states
   e. No flag is affected.

Practice Exercise

1. In an 8085 microprocessor system, the RST instruction will cause an interrupt:
(A) only if an interrupt service routine is not being executed.
(B) only if a bit in the interrupt mask is made 0.
(C) only if interrupts have been enabled by an EI instruction.
(D) None of the above

2. The number of hardware interrupts (which require an external signal to interrupt) present in an 8085 microprocessor are:
(A) 1
(B) 4
(C) 5
(D) 13

3. An 8085 microprocessor responds to the presence of an interrupt:
(A) as soon as the TRAP pin becomes 'high.'
(B) by checking the TRAP pin for 'high' status at the end of each instruction fetch.
(C) by checking the TRAP pin for 'high' status at the end of the execution of each instruction.

(D) by checking the TRAP pin for 'high' status at regular intervals.

4. Identify the software interrupt instruction in an 8085 microprocessor.
(A) INTR
(B) RST 5
(C) RST 7.5
(D) None of the above

5. The number of objects crossing a window sequentially at variable speed is to be counted using an interrupt in the 8085 microprocessor. The objects are sensed by an optical source and a detector with associated signal conditioning circuit. The circuit produces a logical high output as long as the object is in form to the window, and this output is used to interrupt the processor. The duration of an object being in front of the window is in the range of 100ms to 2s. The processor takes 1ms to process the input after an interrupt. The best choice of interrupt for an error-free counting is:
(A) RST 5.5
(B) RST 6.5
(C) RST 7.5
(D) INTR

6. Read the following assembly language program segment of 8085.

```
EI
RIM
ANI 80 H
SIM
```

What kind of task is performed by the above set of instructions?
(A) Sets bit out on SOD pin
(B) Accepts bit in form SID pin
(C) Accepts RST 7.5 interrupt
(D) Resets RST 7.5 interrupt

7. Consider the following program.

```
        ORG 7000H
BEGIN:  LXI H, 7000H
        MOV A,L
        ADD H
        JM END
        RST 0
END:    PCHL
        HLT
```

Which of the following statements is correct?
(A) The program will halt the processor.
(B) The program will be repeated infinitely.
(C) The program will branch to 0007H after JM END.
(D) The program will branch to 0000H after JM END.

8. Match List I (request) with List II (device) and select the correct answer using the codes given below the lists.

List I
a. Interrupt
b. Wait
c. Hold

List II
I. I/O device
II. DMA controller
III. Memory

Codes:

	a	b	c
(A)	I	II	III
(B)	I	III	II
(C)	II	I	III
(D)	II	III	I

9. In a microprocessor when a CPU is interrupted, it:
(A) stops the execution of instructions.
(B) acknowledges the interrupt and branches of subroutine.
(C) acknowledges the interrupt and continues.
(D) acknowledges the interrupt and waits for the next instruction form the interrupting device.

10. The TRAP is one of the interrupts available in INTEL 8085. Which one of the following statements is true of TRAP?
(A) It is level triggered.
(B) It is negative edge triggered.
(C) It is positive edge triggered.
(D) It is both positive edge and level triggered.

11. The highest priority in an 8085 microprocessor system is:
(A) RST 7.5
(B) RST 6.5
(C) INTR
(D) TRAP

12. A fragment of assembly code is given below for 8085.
EI
MVI A, 09
SIM
This sequence of instruction:
(A) enables RST 7.5, RST 6.5, and RST 5.5.
(B) enables RST 7.5 and RST 6.5, and disables RST 5.5.
(C) enables RST 7.5, and disables RST 6.5 and RST 5.5.
(D) disables RST 7.5, RST 6.5, and RST 5.5.

13. For an 8085 microprocessor, the instruction RST 6 restarts subroutine at address:
(A) 0000H
(B) 0030H
(C) 0020H
(D) 0033H

14. Three devices A, B and C are connected to an Intel 8085 A microprocessor. Device A has the highest priority and device C has the lowest priority. The correct assignment of interrupt inputs is:
(A) A uses RST 5.5, B uses RST 6.5, and C uses TRAP.
(B) A uses RST 5.5, B uses RST 6.5, and C uses RST 7.5.
(C) A uses RST 7.5, B uses RST 6.5, and C uses RST 5.5.
(D) A uses TRAP, B uses RST 5.5, and C uses RST 6.5.

15. Match List I (interrupts) with List II (corresponding characteristics) and select the correct answer using the codes given below the lists.
List I
a. TRAP
b. INTR
c. RST 7.5
d. RST 6.5

List II
I. Level triggered
II. Non-maskable
III. For increasing the number of interrupts
IV. Positive edge triggered

Codes:

	a	b	c	d
(A)	II	IV	III	I
(B)	I	IV	III	II
(C)	I	III	IV	II
(D)	II	III	IV	I

16. Which one of the following statements is correct?
In Intel 8085, the interrupt enable flip-flop reset by:
(A) DI instructions only.
(B) system RESET only.
(C) interrupt acknowledgment only.
(D) either DI or system RESET or interrupt acknowledgment.

17. Which one of the following is not a vectored interrupt?
(A) TRAP
(B) INTR
(C) RST 3
(D) RST 7.5

18. Consider the following statements.
The SIM instruction outputs the contents of accumulator to define:
I. interrupt bit
II. interrupt pending bit
III. serial input data lint
IV. serial output data line
Which of the statements given above are correct?
(A) I and II
(B) II and III
(C) III and IV
(D) I and IV

19. Which interrupt has the highest priority?
(A) RST 7.5
(B) RST 7
(C) RST 6.5
(D) INTR

20. Three devices P, Q, and R have to be connected to an 8085 A microprocessor. Device P has the highest priority and device R has the lowest priority. In this context, which of the following is the correct assignment of interrupt inputs?

(A) P uses TRAP, Q uses RST 5.5, and R uses RST 6.5.
(B) P uses RST 5.5, Q uses RST 6.5, and R uses RST 7.5.
(C) P uses RST 7.5, Q uses RST 6.5, and R uses RST 5.5.
(D) P uses RST 5.5, Q uses RST 6.5, and R uses TRAP.

21. When TRAP interrupt is triggered in an Intel 8085A, the program control is transferred to which one of the following?

(A) 0020H
(B) 0024H
(C) 0028H
(D) 002c H

22. What is an interrupt in which the external device supplies its address, as well as, the interrupt request known as?

(A) Vectored interrupt
(B) Maskable interrupt
(C) Non-maskable interrupt
(D) None of the above

23. Which one of the following interrupts is both level and edge sensitive?

(A) RST 7.5
(B) RST 5.5
(C) TRAP
(D) INTR

24. Which value a program counter will have when a non-maskable interrupt of 8085 microprocessor is serviced?

(A) 0004 H
(B) 0014 H
(C) 0024 H
(D) 0034 H

25. Which one of the following has the highest priority?

(A) RST 7.5
(B) RST 5.5
(C) TRAP
(D) HOLD

Answer Key

1. (C)	**2.** (C)	**3.** (A)	**4.** (B)	**5.** (C)	**6.** (C)	**7.** (B)	**8.** (B)	**9.** (B)	**10.** (D)
11. (D)	**12.** (B)	**13.** (B)	**14.** (C)	**15.** (D)	**16.** (D)	**17.** (B)	**18.** (D)	**19.** (A)	**20.** (C)
21. (B)	**22.** (D)	**23.** (C)	**24.** (C)	**25.** (A)					